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Nondestructive evaluation of GFRP composite including multi-delamination using THz spectroscopy and imaging

Dae-Hyun Han, Lae-Hyong Kang*

Department of Mechatronics Engineering and LANL-CBNU Engineering Institute-Korea, Chonbuk National University, 567 Baekje-daero, Deokjin-gu, Jeonju 54896, Republic of Korea

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ABSTRACT

We utilize a fiber-coupled terahertz time-domain spectroscopy system (THz-TDS) based on photoconductive antennas to visualize multi-delamination and their thickness in a glass-fiber-reinforced polymer (GFRP) composite plate. We simulate a hidden multi-delamination by inserting a Teflon film in the GFRP composite plate that is fabricated using the vacuum-assisted resin transfer molding (VARTM) process. Reflected or transmitted terahertz (THz) waves are recorded from GFRP samples mounted on the x–y linear motorized stage during the investigation. The x–y linear motorized stage is driven in steps of 0.2 mm, thereby allowing us to scan the sample. We examine and compare the performances of the THz time-domain visualization (TTV) and THz frequency-domain visualization (TFV) algorithms to evaluate their characteristics for the visualization of defects. The thickness of the GFRP sample and multi-delamination is estimated using the reflection geometry method. Finally, we confirm that the hidden multi-delamination of the GFRP sample has been successfully visualized in the B- and C-scans. In addition, the thicknesses of the GFRP sample and the simulated delamination as compared with the experimental values exhibit a close consistency.

1. Introduction

Composite materials are widely used in fields such as aerospace, wind-energy technology, automobile industry, civil engineering, and ship maintenance due to their mechanical strength, stiffness, corrosion resistance, and low weight. However, the manufacturing of composite structures involves a complex procedure [1–3] in which different types of defects can be generated, for example, delamination, voids, cracks, dry fibers, etc. [4] In addition, defects are also generated and "grown" during the material's lifecycle. Therefore, irregular characteristics that can predictably lead to defects during a composite material's lifecycle should also be considered as defects. Moreover, as the usage of composite materials continues to increase and their application fields continue to extend, their quality and performance specifications are becoming more and more stringent. The main defect in composite materials is delamination, which is the separation of layers due to various stresses [5]. Therefore, it is very important to detect, locate, and visualize hidden delaminations in composite materials.

Many research groups have investigated and developed nondestructive evaluation (NDE) or nondestructive testing (NDT) methods [4], such as computed tomography (CT) [6,7], X-ray testing [8], liquid penetrant testing (PT) [9], magnetic particle testing (MT) [10], electromagnetic testing (ET), thermal infrared testing (IR) [11], and ultrasound testing (UT) [12,13] in order to detect hidden delaminations. Each technique has its own advantages and disadvantages. CT is limited to small-sized test specimen due to the structural limitations of the equipment and high cost of installation and maintenance. Meanwhile, during X-ray testing, composite specimens may be damaged, and operator safety may be compromised due to high radiation energy. Further, PT and MT can only detect surface cracks. While IR requires a thermal source to improve performance, composite materials will undergo thermal deformation when exposed to heat sources over long periods of time. UT requires a liquid couplant to efficiently transmit ultrasound signals to the specimen. In addition, UT can estimate the location of a defect, but the depth of the defect from the surface is difficult to detect, because ultrasonic waves are largely attenuated in air, including internal air layers formed due to delamination.

Recently, terahertz (THz) imaging technology has become the most promising technique for NDT [14]. The THz part of the electromagnetic spectrum extends from 100 GHz to 10 THz, lying between the microwave and infrared regions, and the wavelength range in this region spans from 3 mm down to 30 μm, as shown in Fig. 1. THz waves are not harmful to biological tissue, and they can provide high spatial resolution due to their shorter wavelengths. Previously, it was difficult to

* Corresponding author.
E-mail address: reon.kang@jbnu.ac.kr (L.-H. Kang).

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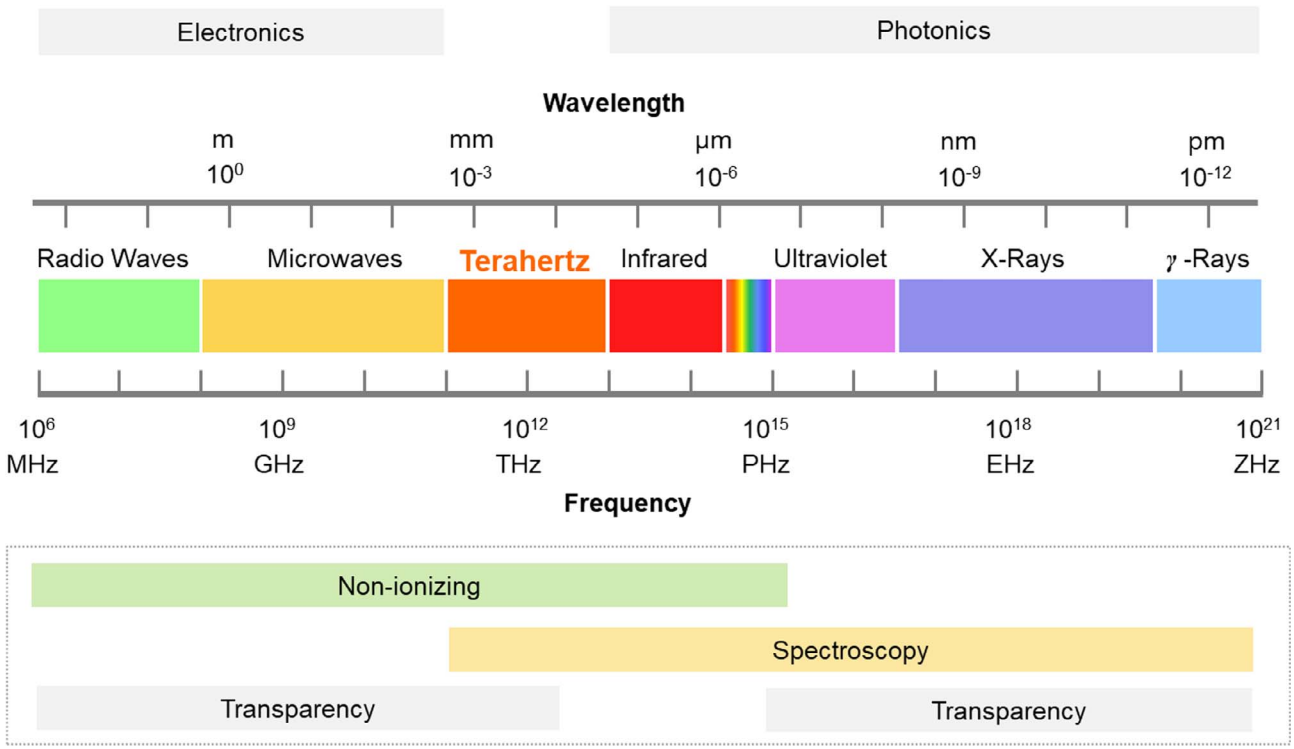

Fig. 1. THz band in the electromagnetic spectrum.

implement THz imaging due to the technical limitations of efficiently generating and detecting THz waves. However, the development of ultrafast lasers and ultra-micro machining technologies has contributed to the development of THz time-domain spectroscopy (THz-TDS). This evolution has enabled the introduction of THz-based technology to NDT, making apparent its advantages for the inspection of composite materials.

Many studies on THz-based inspection and visualization of defects in composite structures have been actively carried out to improve on the existing NDT limitations. Furthermore, THz-based inspection methods do not require a liquid couplant or any additional parts during inspection.

Table 1 summarizes the THz-based visualization methods proposed thus far. Redo-Sanchez et al. [15] demonstrated that occluded textual content could be successfully extracted from a packed stack of paper pages down to nine pages without human supervision via the use of a fiber-coupled pulsed THz-TDS system based on a photoconductive antenna (PCA). THz images were captured by mounting samples on an x–y motorized stage with 0.2-mm step size. Meanwhile, Dong et al. [16] considered reflection and transmission methods using a THz-TDS system based on PCAs to visualize a Teflon film in a glass-fiber-reinforced polymer (GFRP) plate consisting of eight glass fiber layers. Ryu et al. [17] visualized a simulated delamination in a GFRP plate by using a fiber-coupled pulsed THz-TDS system based on PCA. They set the incident angle to 25° in the reflection method. Further, Zhang et al. [18] attempted to image a Teflon film in a carbon-fiber-reinforced polymer (CFRP) plate consisting of two carbon fibers. They considered reflection methods using a THz-TDS system based on PCAs and processed the THz signals in the time domain. Sørgård et al. [19] studied a CFRP plate fabricated with twelve carbon fibers that included a simulated delamination between the carbon fiber layers. However, in all these studies, defect visualization was performed in either the time domain or the

Table 1
Summary of existing defect visualization methods.

	THz Source	THz band width (THz)	Time delay range (ps)	Scan size (mm × mm)	Scan step (mm)	Scan geometry
	Incident angle	Specimen	Defect type (Thickness)	Imaging method	Humidity	Scan method
Redo-Sanchez et al. [15]	Fiber-coupled PCA, Pulsed THz	0–2	0–100	22 × 44	0.2	Reflection
	–	9 layers paper	Text (20 μm)	Time-gated spectral, Time domain amplitude	–	Raster scan
Dong et al. [16]	PCA, Pulsed THz	0.6–3	0–45	50 × 50	0.1	Reflection, transmission
	–	8 layers GFRP	Teflon film (250 μm)	Time domain amplitude	–	Raster scan
Ryu et al. [17]	Fiber-coupled PCA, Pulsed THz	0.1–3	26–48	30 × 30	–	Reflection
	–	12 layers GFRP	Delamination (100 μm)	Time domain amplitude	1%	Raster scan
Zhang et al. [18]	PCA, Pulsed THZ	0–3.5	0–80	100 × 100	–	Reflection
	–	2 layers CFRP	Teflon film (150 μm)	Time domain amplitude	–	Raster scan
Sørgård et al. [19]	Fiber-coupled PCA, Pulsed THz	–	80–300	60 × 60	1	Reflection, transmission
	–	12 layers CFRP	Impact damage (-)	Peak-to-peak, Time domain amplitude	–	Raster scan

frequency domain with a small scan area.

In this study, we use a fiber-coupled PCA-based THz-TDS system pumped by an 80-fs fiber laser to visualize delamination in the transmission and reflection modes. THz images are acquired by mounting the samples on an x–y stage, which is stepped through the area of the beam while the transmitted or reflected THz waveform is captured. The GFRP plate is fabricated using vacuum-assisted resin transfer molding (VARTM) in the lab, and it includes an artificially fabricated delamination between the glass-fiber layers. The results of THz-TDS and THz frequency-domain spectroscopy (THz-FDS) are compared with the experimental results. The THz-TDS technique is used to measure the thickness and location of each delamination along the z-direction, while THz-FDS evaluates the defect size and fiber direction according to the frequency range. In addition, the peak time shift algorithm is utilized to indicate the defect location in the large scan area. Finally, the THz-TDS and THz-FDS imaging results obtained with several algorithms are compared.

2. Generation and detection of pulsed THz waves

The application of a PCA, optical rectification, or the surface electric field of a semiconductor can be used to generate pulsed THz waves. These methods are based on the nonlinear characteristics of materials, and the methods can generate the THz waves with a wide bandwidth from GHz to THz [20]. THz-wave generation and detection schemes are classified as either coherent or incoherent techniques. The fundamental difference is that coherent detection measures both the amplitude and phase of the field, whereas incoherent detection measures only the intensity. Coherent detection techniques are closely associated with generation techniques in that they share the underlying mechanisms and key components. In particular, the optical techniques for THz radiation and detection utilize the same light source.

2.1. Pulsed THz-wave generation using PCA

Pulsed THz waves can be generated from a biased PC antenna excited by femtosecond laser pulses. Fig. 2 illustrates pulsed THz-wave emission from a commonly used THz emitter structure. The THz emitter comprises two metal electrodes deposited on a semiconductor substrate. A DC bias is applied between the electrodes. Femtosecond optical pulses with photon energy larger than the bandgap of the semiconductor generate free electron–hole pairs in the gap between the electrodes. The static bias field accelerates the free carriers, and simultaneously, the charge density declines primarily via trapping of carriers in defect sites on the time scale of carrier lifetimes. The impulse current arising from the acceleration and decay of free carriers forms the source of the pulses of electromagnetic radiation.

2.2. Measurement of pulsed THz waves using PCA

It is not easy to measure THz waves at room temperature directly because their energy is very small. However, technological advances in optics and electronics have made it possible to measure THz waves at room temperature. In the special case of pulsed THz waves, the general method for measurement is the electro-optic (EO) sampling method. Free-space EO sampling measures the actual electric field of broadband THz pulses in the time domain by utilizing the Pockels effect, which is closely related to optical rectification. A THz field induces birefringence in a nonlinear optical crystal, which is proportional to the field amplitude. The entire THz waveform is determined by a weak optical probe measuring the field-induced birefringence as a function of the relative time delay between the THz and optical pulses. Sensing with a PC antenna also allows the measurement of broadband THz pulses in the time domain. In the absence of a DC bias field, the THz field induces a current in the photoconductive gap when an optical probe pulse injects photocarriers. The induced photocurrent is proportional to the THz field amplitude. The THz pulse shape is mapped out in the time domain via measurement of the photocurrent while varying the time delay between the THz pulse and the optical probe.

Fig. 3 shows the schematic of the time-resolved measurement of pulsed THz waves with the use of a PCA receiver. The pulsed THz waves can be measured using the inverse process of pulsed-THz-wave generation using a PCA. Carriers are generated when a femtosecond laser beam is irradiated between both electrodes of the PCA. During this time, the pulsed THz waves that have arrived at the PCA accelerate the generated carriers, causing the electric field of the pulsed THz waves to generate a fine current pulse. This current is subsequently measured through a current meter installed at the electrodes. The electric field strength of the pulsed THz waves can be indirectly measured because the current value is proportional to the amplitude of the pulsed THz waves.

3. Theoretical background for analyzing THz signals

3.1. Thickness of sample or defect

The time-resolved detection scheme of THz-TDS is directly applicable to measuring the depth information of a multi-layer sample. When pulsed THz waves are incident on an object, the reflected THz waveform consists of a series of pulses reflected from the interfaces [21]. The thickness of the sample or defect can be calculated in the THz reflection scan mode via the following equation:

$$T_{sample} = \frac{\Delta t}{2} \times \frac{c}{n_{sample}} \times \cos\theta, \tag{1}$$

where T_{sample} represents thickness of the sample, Δt the time between

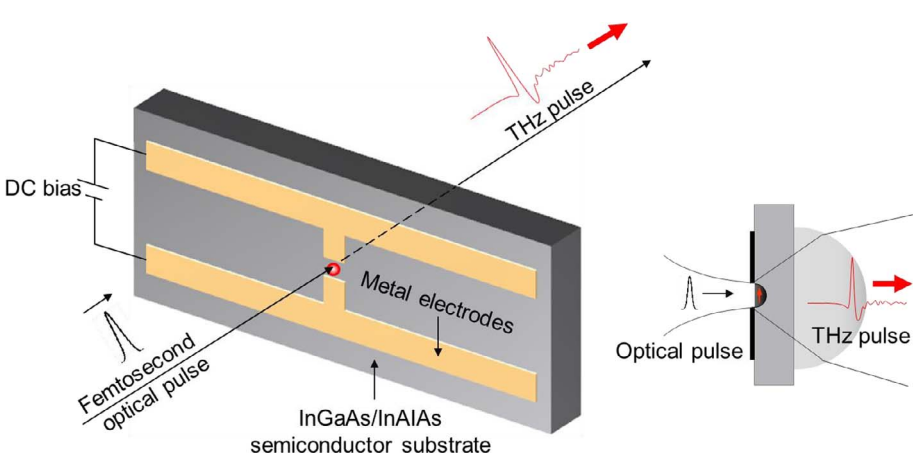

Fig. 2. Schematic of THz pulse emission using photoconductive antenna excited by a femtosecond laser pulse.

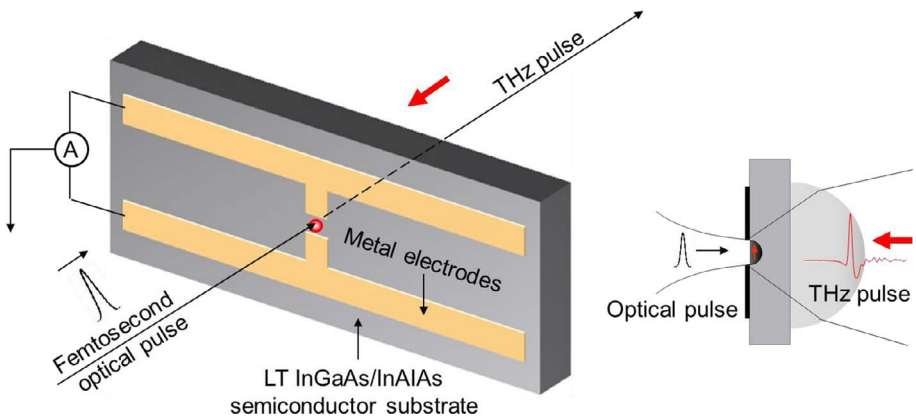

Fig. 3. Schematic of THz pulse detection using photoconductive antenna excited by a femtosecond laser pulse.

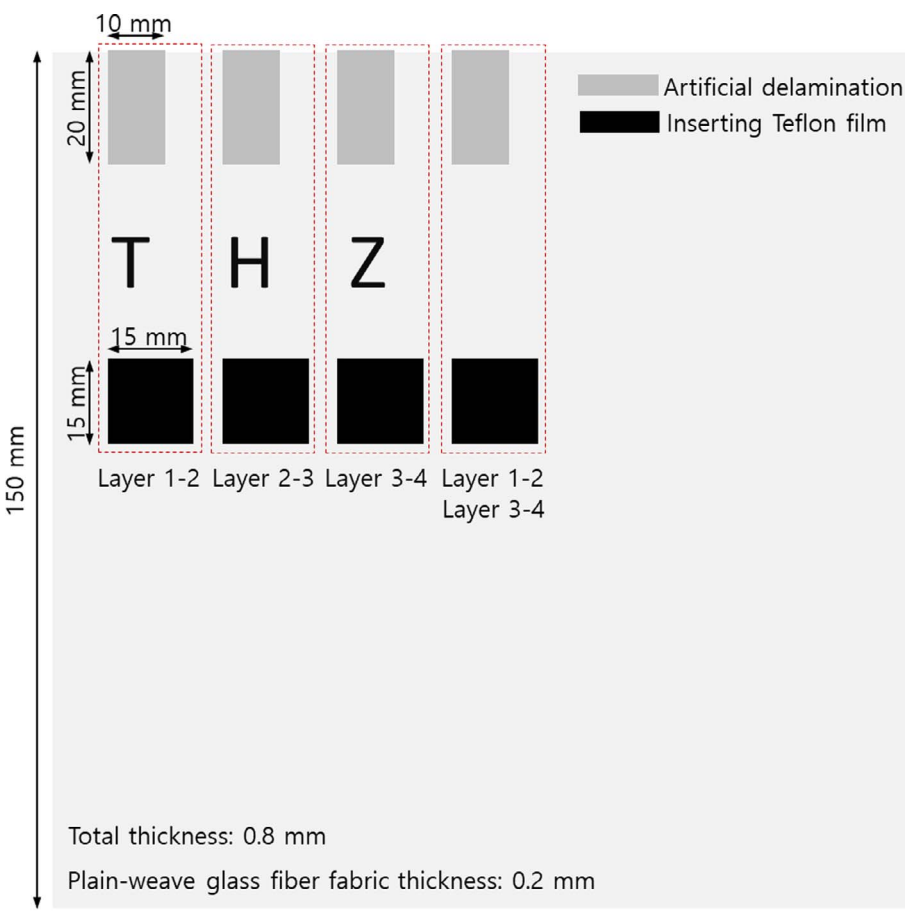

Fig. 4. Schematic of glass-fiber-reinforced polymer (GFRP) sample with inserted Teflon film and artificial delamination.

successive reflections, c the speed of light in air, n_{sample} the refractive index of sample, and θ the incident angle of the THz waves. The factor of one-half arises since the THz waves are measured in the reflection mode [16]. The refractive index (n) of common GFRPs is known to be 2.1 [22] while that of common Teflon and air are 1, respectively [23].

3.2. Fraction of incident THz power

The fraction of the incident THz power that is reflected from the interface is given by the reflectance. In order to evaluate the path of the THz waves, the reflection and transmission power of THz waves at the interface of two media (say 1 and 2) can be calculated using the Fresnel equations as follows [24]:

$$R_s = \left|\frac{Z_2\cos\theta_i - Z_1\cos\theta_t}{Z_2\cos\theta_i + Z_1\cos\theta_t}\right|^2 = \left|\frac{\sqrt{\frac{\mu_2}{\varepsilon_2}}\cos\theta_i - \sqrt{\frac{\mu_1}{\varepsilon_1}}\cos\theta_t}{\sqrt{\frac{\mu_2}{\varepsilon_2}}\cos\theta_i + \sqrt{\frac{\mu_1}{\varepsilon_1}}\cos\theta_t}\right|^2 \tag{2}$$

$$R_p = \left|\frac{Z_2\cos\theta_t - Z_1\cos\theta_i}{Z_2\cos\theta_t + Z_1\cos\theta_i}\right|^2 = \left|\frac{\sqrt{\frac{\mu_2}{\varepsilon_2}}\cos\theta_t - \sqrt{\frac{\mu_1}{\varepsilon_1}}\cos\theta_i}{\sqrt{\frac{\mu_2}{\varepsilon_2}}\cos\theta_t + \sqrt{\frac{\mu_1}{\varepsilon_1}}\cos\theta_i}\right|^2, \tag{3}$$

where R_s represents the reflectance of s-polarized THz waves, R_p the reflectance of p-polarized THz waves, Z_1 and Z_2 the wave impedances of the media involved (media 1 and 2), μ_1 and μ_2 the magnetic permeabilities of materials 1 and 2, respectively, ε_1 and ε_2 the electric permittivities of materials 1 and 2, respectively, at the frequency of the THz waves, θ_i the incident THz-wave angle, θ_r the reflectance THz-wave angle, and θ_t the reflection THz-wave angle. In case of non-magnetic

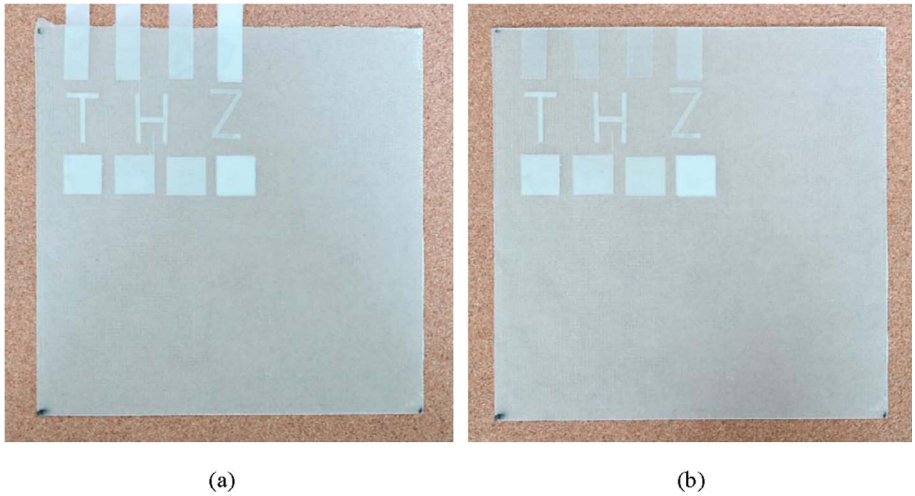

Fig. 5. Actual glass-fiber-reinforced polymer (GFRP) sample with a Teflon artificial insert and artificial delamination: (a) before removing Teflon film; (b) after removing Teflon film to create artificial delaminations.

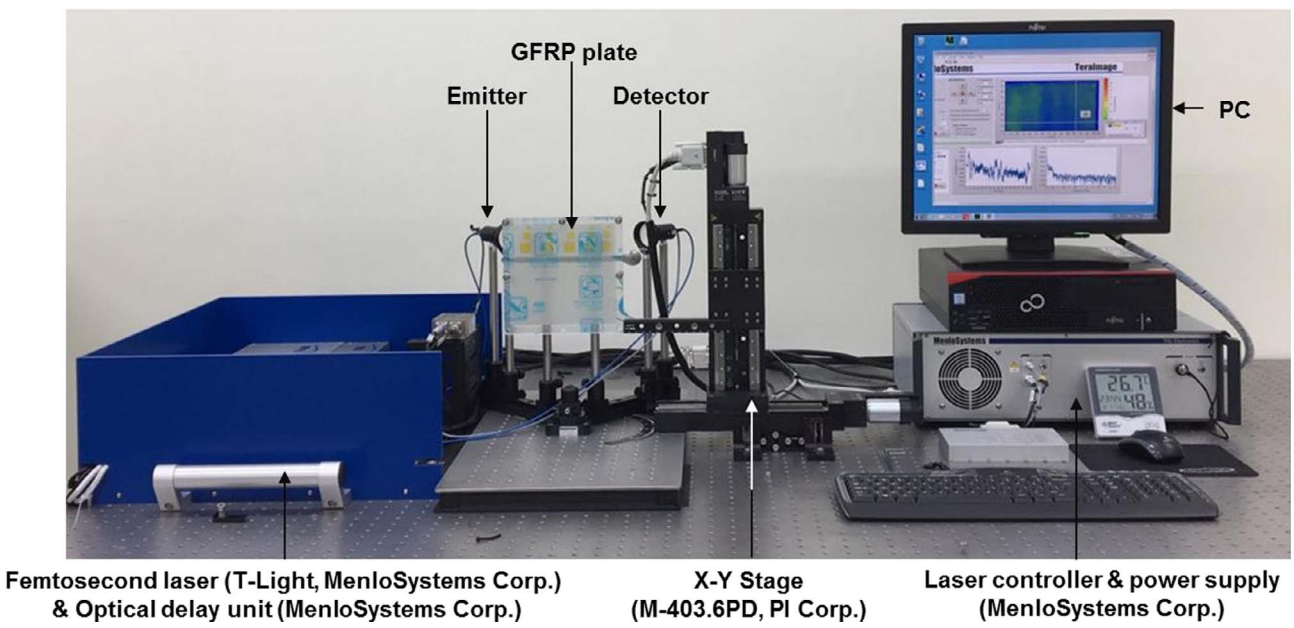

Fig. 6. THz time-domain spectroscopy (THz-TDS) system utilized for defect visualization.

media (i.e., materials for which $\mu_1 \approx \mu_2 \approx \mu_0$, where μ_0 denotes the permeability of free space), impedances Z_1 and Z_2 can be expressed as follows:

$$Z_1 = \frac{Z_0}{n_1}, Z_2 = \frac{Z_0}{n_2}. \tag{4}$$

The second form of Eqs. (5) and (6) can be derived from the first by translating θ_t to θ_i using Snell's law. Thus, the reflectance coefficient for s-polarized THz waves becomes

$$R_s = \left| \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t} \right|^2 = \left| \frac{n_1\cos\theta_i - n_2\sqrt{1-(\frac{n_1}{n_2}\sin\theta_i)^2}}{n_1\cos\theta_i + n_2\sqrt{1-(\frac{n_1}{n_2}\sin\theta_i)^2}} \right|^2, \tag{5}$$

while the reflectance for p-polarized THz waves becomes

$$R_p = \left| \frac{n_1\cos\theta_t - n_2\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_i} \right|^2 = \left| \frac{n_1\sqrt{1-(\frac{n_1}{n_2}\sin\theta_i)^2} - n_2\cos\theta_i}{n_1\sqrt{1-(\frac{n_1}{n_2}\sin\theta_i)^2} + n_2\cos\theta_i} \right|^2. \tag{6}$$

Consequently, the total reflectance (R) of the THz waves can be written as

$$R = R_s + R_p/2, \tag{7}$$

and the total transmittance (T) of the THz waves can be expressed as

$$T = 1 - R. \tag{8}$$

We can calculate the THz-wave power from Eq. (8), corresponding to THz-wave transmission at the GFRP surface.

3.3. Absorbance of THz waves in refraction material

The absorbance of THz waves is the ratio that measures the attenuation of the transmitted THz-wave power in a material. The THz waves incident on the GFRP sample reach the interface between the GFRP and the upper side of the defect, and a portion of THz waves is reflected at the upper side of the defect and transmitted again into the GFRP. The remaining THz waves are transmitted into the defect. When the THz waves pass through the specimen, transmission and reflection are repeated at the interfaces between two different media. The absorbed power of the THz waves can be calculated using Eq. (9) as follows [17]:

$$A = I(1 - e^{-\alpha x}), \tag{9}$$

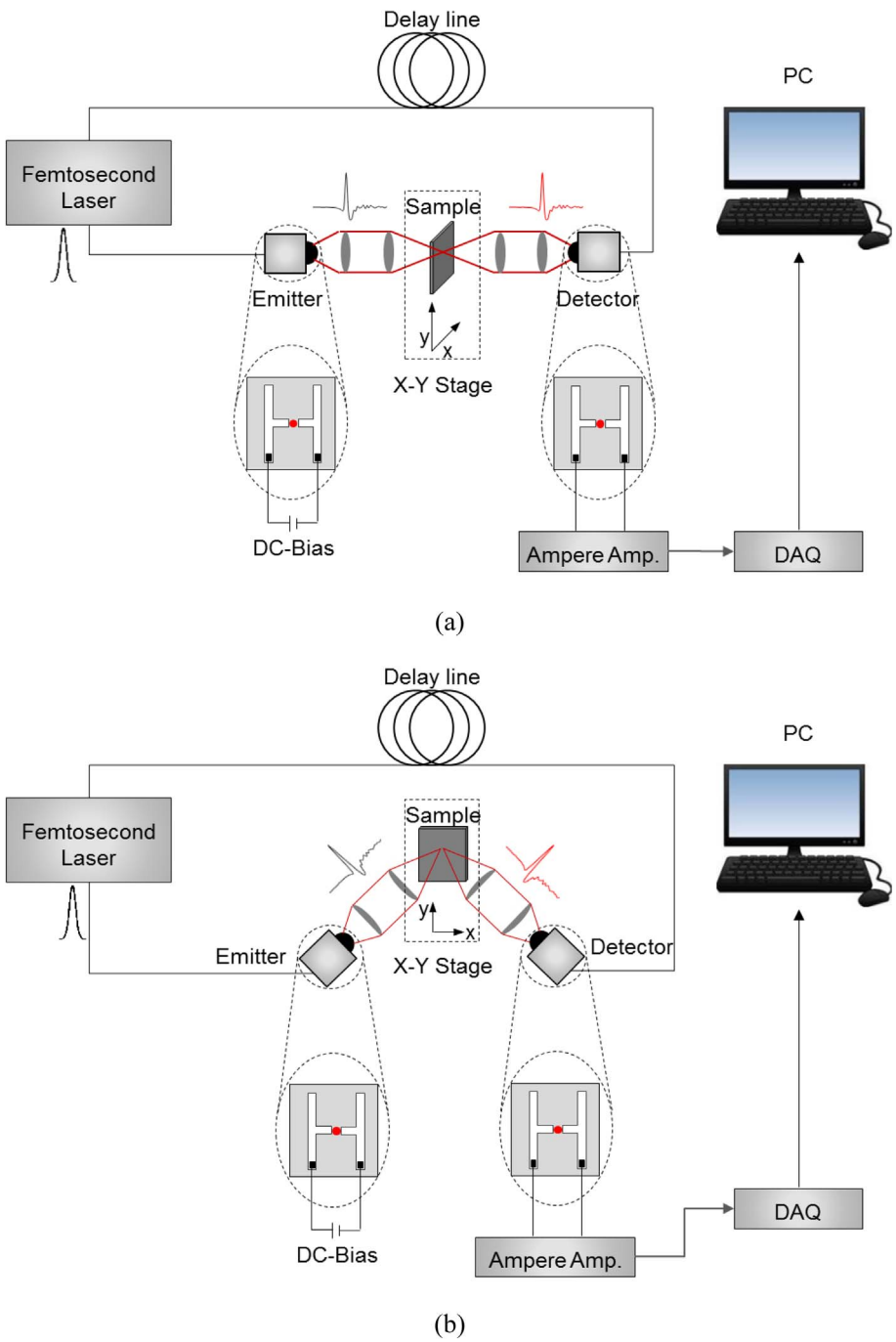

Fig. 7. Schematic of THz time-domain spectroscopy (THz-TDS) system: (a) transmission geometry; (b) reflection geometry.

where A represents the absorbance of the THz waves in the refraction material, I the incident THz-wave power, α the absorption coefficient of the refractive material, and x the thickness of the sample consisting of the same material.

4. Experimental setup

4.1. Fabrication of GFRP specimen

We performed an overview of literature to determine the usual thickness of simulated delamination in composite materials. The simulated delamination thickness range is normally varied from 0.02 mm to 0.2 mm in the literature studies for the visualization of the delamination. Therefore, we finally decided to fabricate the simulated delamination using Teflon film with a thickness of 0.150 mm. The thickness of the simulated delamination in the literature studies is also listed in Table 1.

Fig. 4 shows the GFRP specimen information including the defect type and locations in terms of the depth. Four plain-weave glass fiber fabrics with a thickness of 0.2 mm were stacked to fabricate our square-type GFRP specimen with a side length of 150 mm and thickness of 0.8 mm.

The hidden multi-delamination was simulated by inserting 100 μm thick Teflon films between the plain-weave glass fiber fabrics. The epoxy resin was impregnated into the stacked plain-weave glass fiber fabric by using the VARTM method. The GFRP specimen was cured at 80 °C for 12 h on the electric heating table. We simulated two types of defects to compare their characteristics. The first defect was fabricated by inserting the Teflon film between the plain-weave glass fiber fabrics, while the second defect comprised an artificial delamination that was

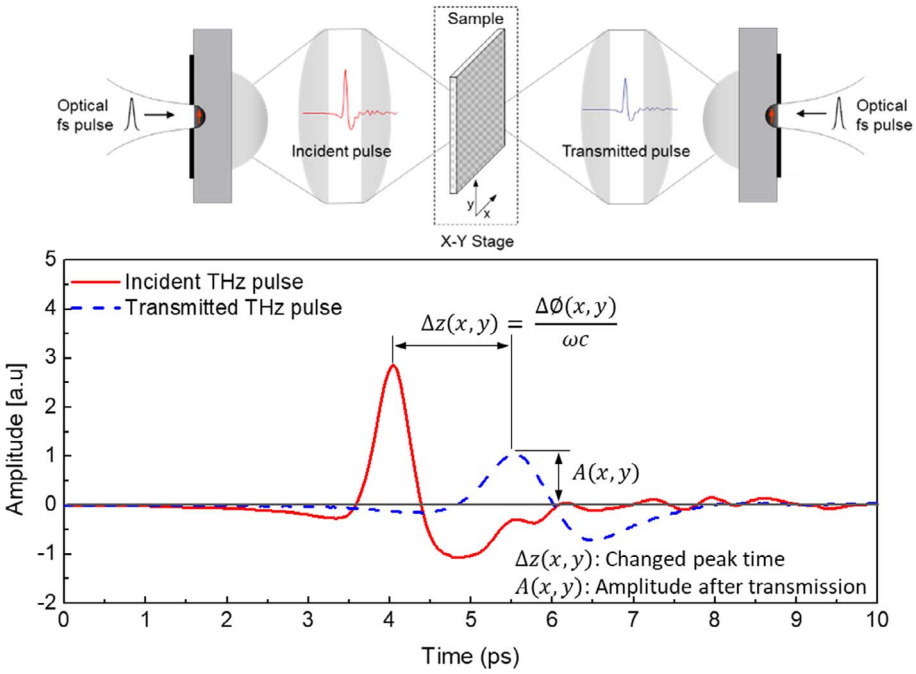

Fig. 8. Schematic of raster scan using THz time-domain spectroscopy (THz-TDS) system along with output waveform.

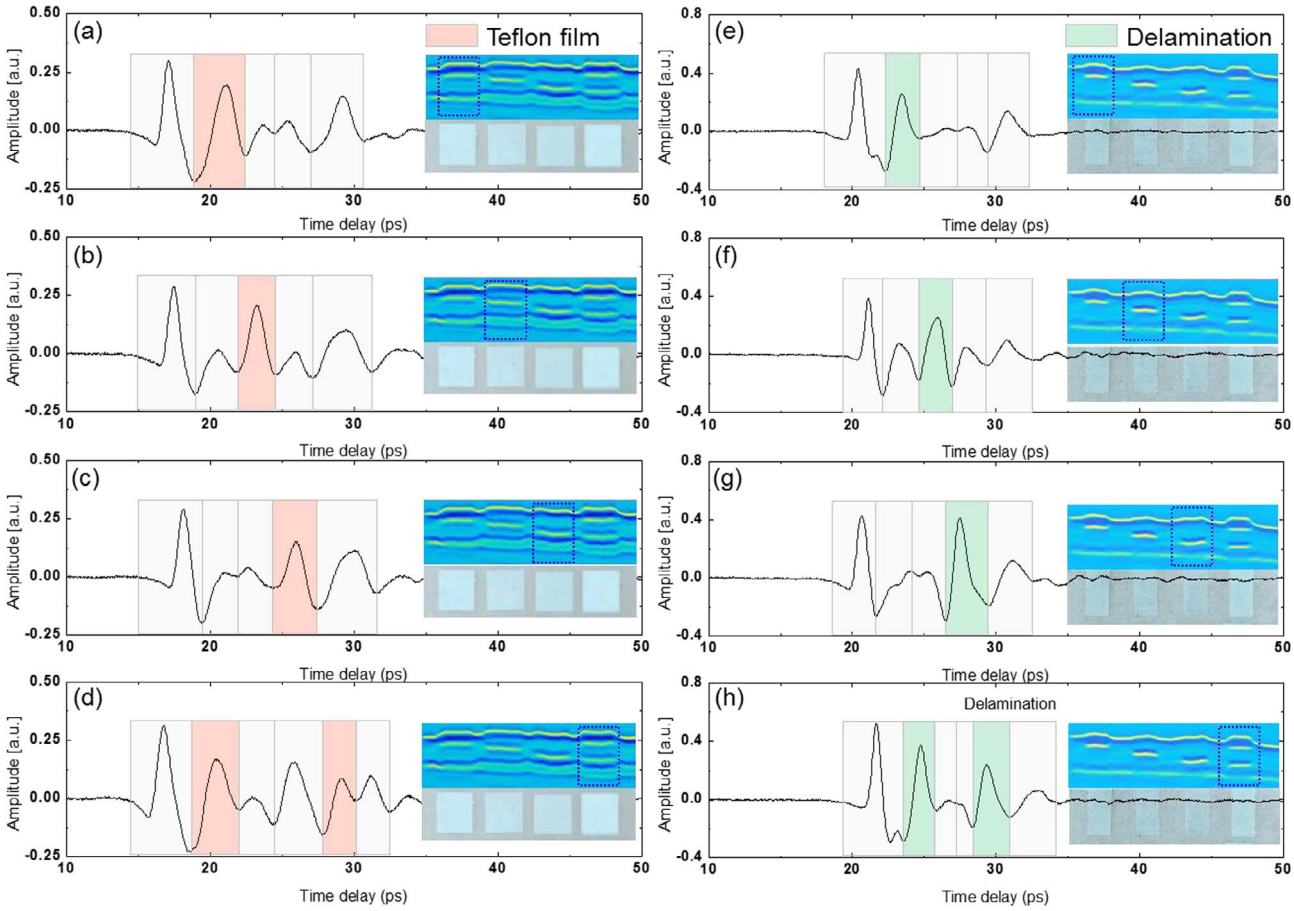

Fig. 9. Time traces of detected reflected THz signal: (a–d) with inserted Teflon film; (e–h) with simulated delamination. The waveforms at the boundaries between the white rectangular boxes in each figure indicate the reflections associated with interfaces between the layer of the glass-fiber-reinforced polymer (GFRP) sample and those of the colored square boxes indicate reflections corresponding with the Teflon film and delamination.

Table 2
Comparison of our results with calculated and actual thickness values.

	Layer thickness (Cal.)							Unit: μm Total thickness	
	Layer 1	Defect 1	Layer 2	Defect 2	Layer 3	Defect 3	Layer 4	Cal.	Act.
GFRP + Teflon film	246	261	100	–	121	–	179	907	893
	241	–	139	203	123	–	154	860	876
	223	–	119	–	121	210	162	835	859
	198	165	113	–	169	178	110	928	963
GFRP + Simulated delamination	264	173	133	–	101	–	146	817	827
	236	–	124	171	116	–	156	793	813
	203	–	113	–	123	227	149	815	821
	256	173	60	–	61	197	154	901	914

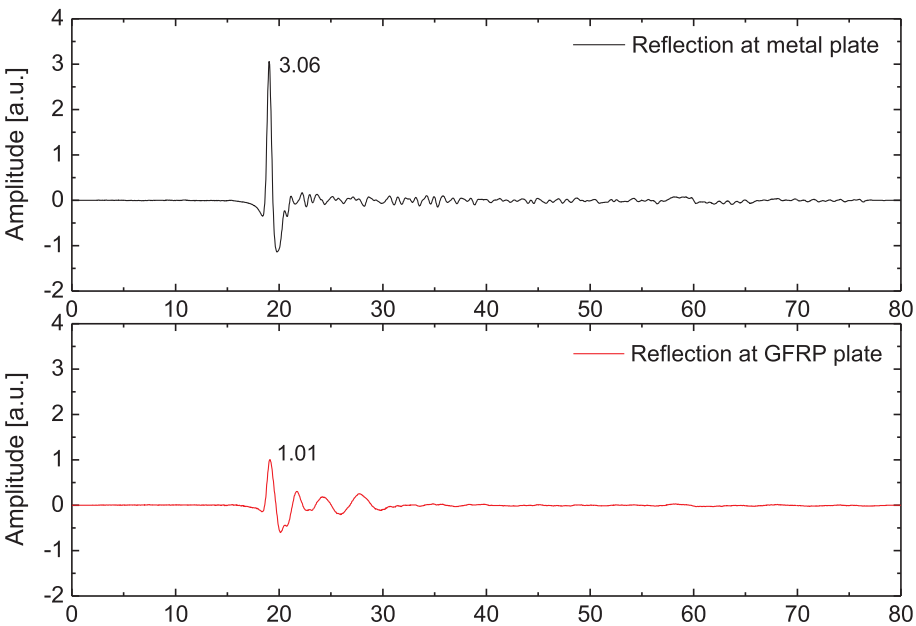

Fig. 10. Comparison of pulsed THz waves in reflection geometry.

fabricated by removing the Teflon film after curing process, as shown in Fig. 5.

4.2. THz spectroscopy imaging system

We employed a commercial fiber-coupled THz spectroscopy system (TERA K15, MenloSystems Corp.) to detect the hidden multi-delamination in the GFRP composite structure. This THz-TDS system has a bandwidth ranging from 0.1 THz to 4 THz. The dynamic range is guaranteed to be > 75 dB, while the total scan range is > 850 ps. For our study, we set the scan range to 80 ps. The THz-TDS system provides an average-THz-wave power output of over 60 mW.

The main components of the TERA K15 system are the femtosecond laser (T-Light, Menlo Systems Corp.), optical delay unit (ODU, MenloSystems Corp.) module, PCAs for the emitter (TERA 15-TX-FC, MenloSystems Corp.) and detector (TERA 15-RX-FC, MenloSystems Corp.), and motorized x–y linear stage (M-403.6PD, PI Corp.), as shown in Fig. 6. The femtosecond laser is an erbium-doped fiber laser that emits pulses with wavelengths of 1560 nm ± 20 nm with 90-fs pulse duration at a 100-MHz repetition rate. The laser has an average output power of over 120 mW and pulse energy of over 1.2 nJ. The ODU is operated at 10 Hz, meaning that the coherent detection of the THz waves is performed 10 times per second by varying the optical path length. The PCA for the emitter consists of an InGaAs/InAlAs semiconductor, and a 100 V DC bias is applied to generate the THz waves. The PCA for the detector consists of a low temperature (LT) InGaAs/InAlAs semiconductor, and an ampere meter is connected to the detector to detect THz waves. The motorized x–y linear stage can scan a maximum area with dimensions of 150 mm × 150 mm with a resolution of < 500 μm. The schematic of the THz-TDS system for transmission and reflection is illustrated in Fig. 7.

The GFRP sample was raster-scanned by a set of motorized x–y moving stages with a 0.2-mm step resolution over a 150 mm × 150 mm area containing the simulated defects, corresponding to 750 × 750 pixels. Each scanned THz waveform contained 2400 data points, corresponding to 1 shot per pixel.

5. Experimental results

The scanning of the GFRP, including the hidden delamination, was performed in a laboratory with the temperature controlled at around 20–22 °C. The humidity in the laboratory was around 46–51%. First, we analyzed the raw THz signal corresponding to the various defects using the transmission and reflection methods to calculate the defect thickness and depth.

5.1. THz spectroscopy results

5.1.1. Transmission geometry scan results

Fig. 8 shows the schematic of the pulsed THz waves in the

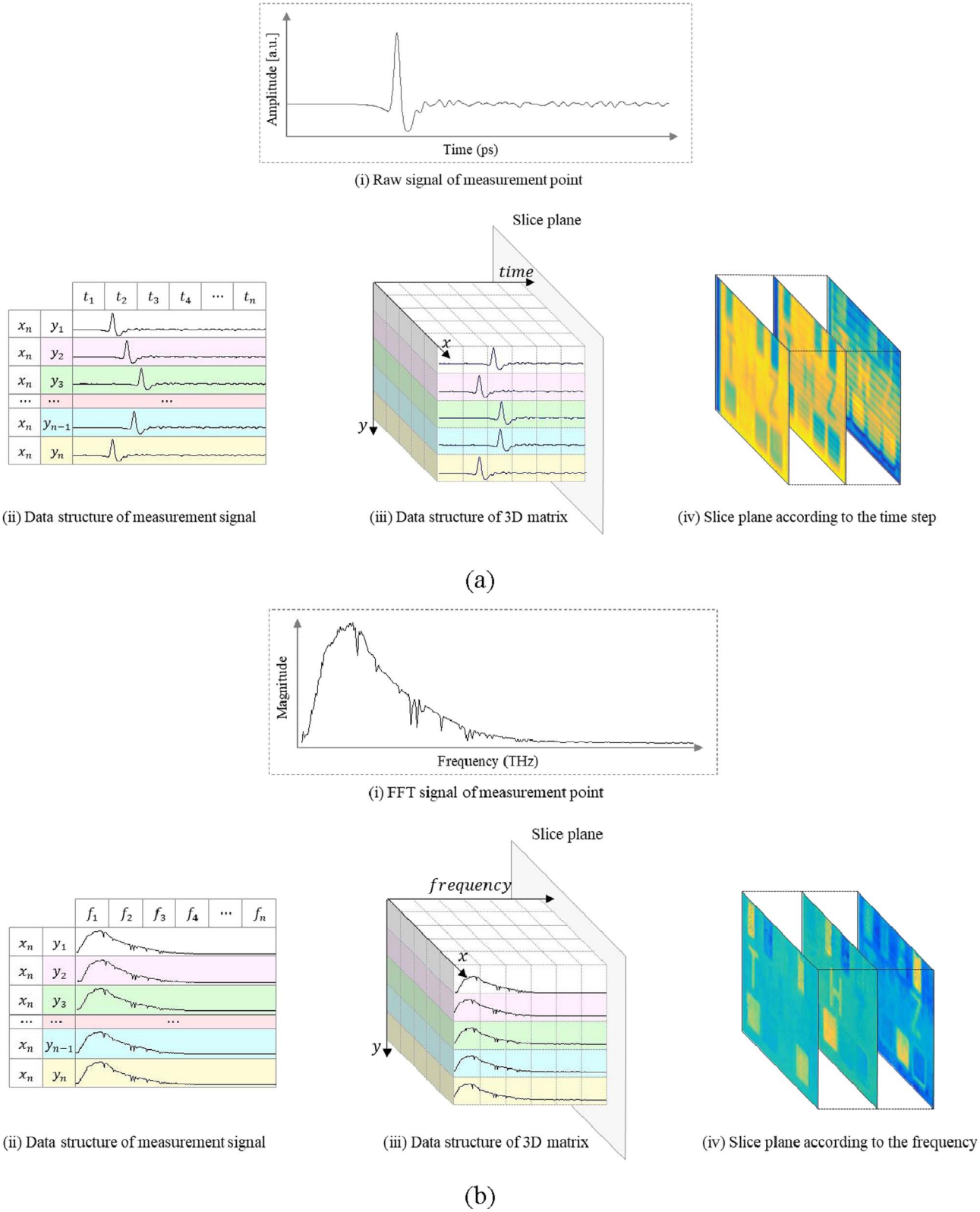

Fig. 11. Image visualization process using THz waveform: (a) THz time-domain visualization (TTV) algorithm; (b) THz frequency-domain visualization (TFV) algorithm.

transmission geometry. The target signal is obtained by analyzing the transmitted THz waveforms. When the pulsed THz waves is transmitted through the materials, the amplitude A(x, y) and the phase ∅ (x, y) change according to material properties such as the refractive index n (x, y), absorption coefficient α (x, y), and thickness d(x, y) [21]. In our study, then the THz waves were transmitted through normal (defect-free) GFRP, the amplitude was 1.06 [a.u.] at 5.48 *ps*. When the THz waves were transmitted in air, the amplitude was 3.02 [a.u.] at 3.97 *ps*. In the time domain, the maximum amplitude (A) or shifted phase (∅) of the waveform is commonly used to visualize the two-dimensional (2D) image.

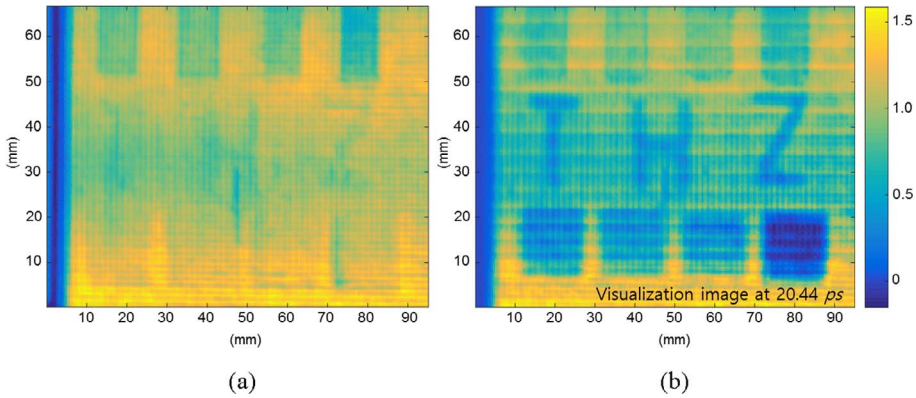

Fig. 12. THz transmission C-scan visualization images: (a) peak-to-peak amplitude method; (b) THz time-domain visualization (TTV) algorithm method.

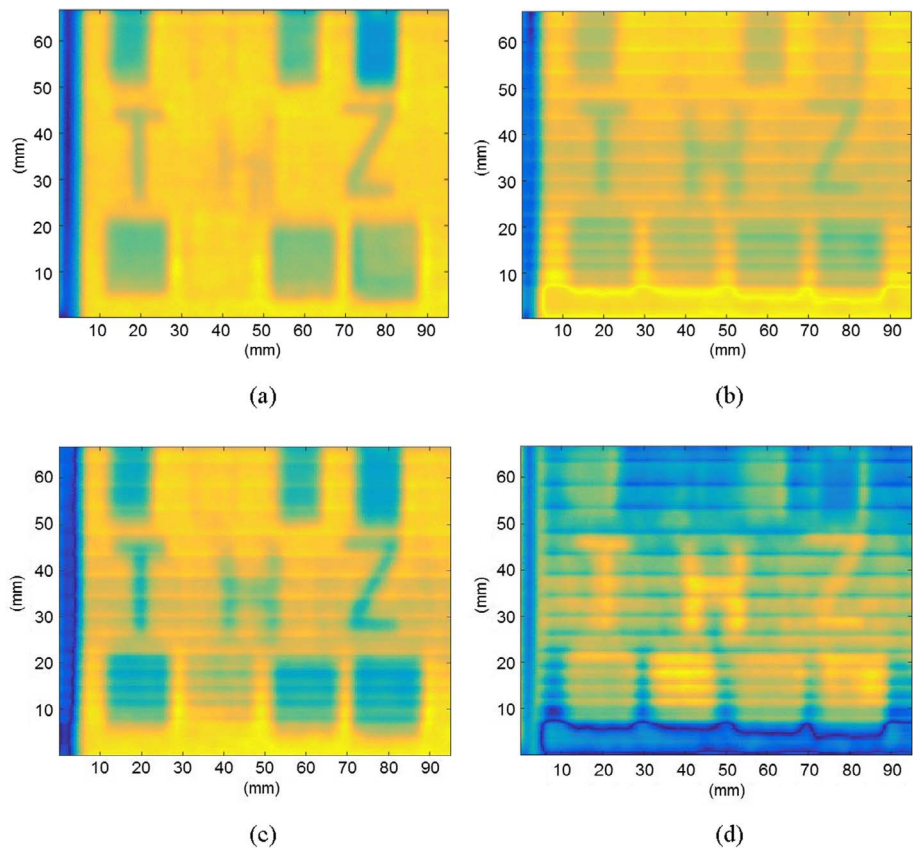

Fig. 13. THz transmission C-scan visualization images obtained using THz frequency-domain visualization (TFV) at 0.225 THz: (a) magnitude map; (b) phase map; (c) real-component map; (d) imaginary-component map.

5.1.2. Reflection geometry scan results

We calculated the thickness of the sample or defect using the reflection geometry scan method. First, the thickness of the sample or defect was calculated using Eq. (1). The thickness of the GFRP can be expressed as

$$T_{GFRP} = \frac{\Delta T}{2} \times \frac{c}{n_{GFRP}} \times \cos\theta = \frac{12.105ps}{2} \times \frac{300 \times 10^6 m/s}{2.16} \times \cos 43.2° \approx 650\mu m, \tag{10}$$

where c denotes the speed of light, ΔT the time difference value between first peak and last peak, and θ the incident angle of the THz waves. The calculated thickness of the GFRP is around 650 μm, close to the actual thickness of 677 μm measured using micrometer. The thickness of the Teflon film located between first and second layers can be expressed as

$$T_{Teflonfilm} = \frac{\Delta T}{2} \times \frac{c}{n_{Teflon}} \times \cos\theta = \frac{2.104ps}{2} \times \frac{300 \times 10^6 m/s}{1.4} \times \cos 43.2° \approx 160\mu m, \tag{11}$$

and this thickness value is also close to the of actual Teflon-film thickness value of 150 μm. The thickness of the artificially simulated delamination can be expressed as

$$T_{dela.} = \frac{\Delta T}{2} \times \frac{c}{n_{dela.}} \times \cos\theta = \frac{1.511ps}{2} \times \frac{300 \times 10^6 m/s}{2.1} \times \cos 43.2° \approx 161\mu m, \tag{12}$$

as shown in Fig. 9, and the measured thickness using a micro-meter in each case is listed in Table 2.

Second, the transmitted intensity of the THz waves can be calculated using Eq. (8), where θ_i is 43.2°, n_{air} is around 1, and n_{GFRP} is around 2.16. THz waves were transmitted into the GFRP with 68.1% of the waves transmitted through the surface, while 31.9% of the waves were reflected. This reflected THz component was captured as the

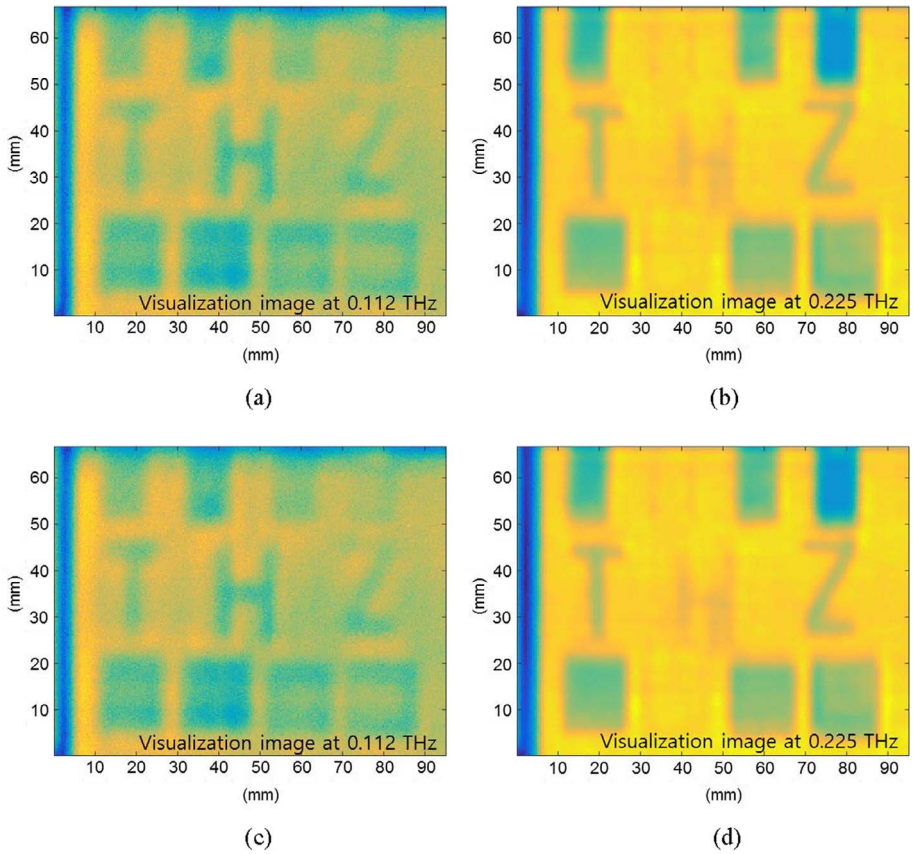

Fig. 14. THz transmission C-scan visualization images obtained at different frequencies: (a) 0.112 THz; (b) 0.225 THz; (c) 0.399 THz; (d) 0.862 THz.

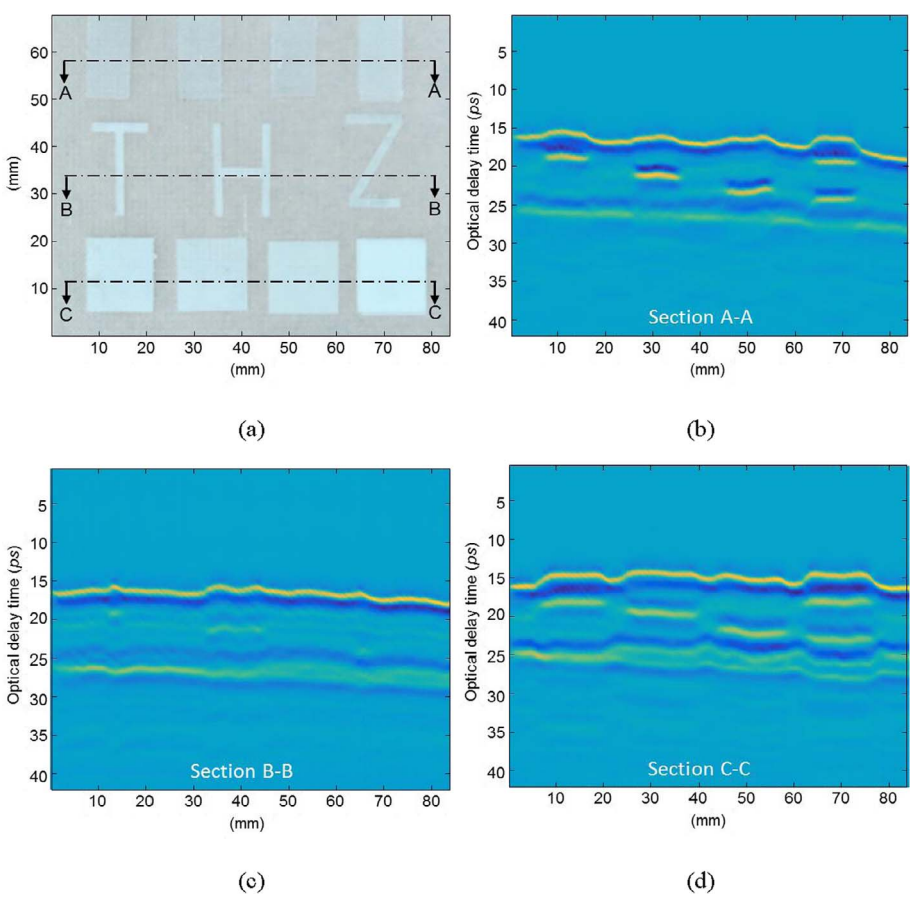

Fig. 15. THz reflection B-scan visualization images obtained using THz time-domain visualization (TTV): (a) actual image; (b) cross-section A-A; (c) cross-section B-B; (d) cross section C-C.

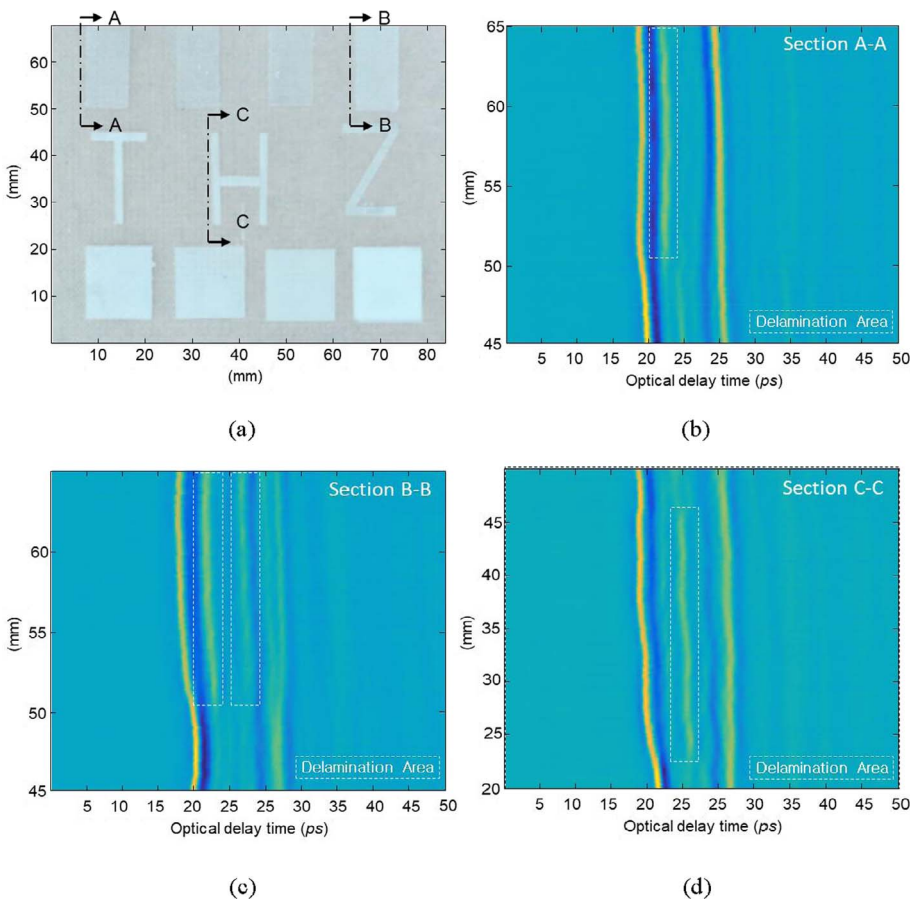

Fig. 16. THz reflection B-scan visualization images obtained at the boundary of artificial delamination using THz time-domain visualization (TTV): (a) actual image; (b) cross-section A-A; (c) cross-section B-B; (d) cross section C-C.

intensity of reflected THz. The first peak intensity value of the THz waves from the emitter was 3.06, and the first peak intensity value of the reflected THz waves from the GFRP surface was 1.01. Both signals are compared in Fig. 10.

5.2. THz imaging results

The B-scan is a diagnostic method to generate a 2D, cross-sectional view of the sample. A THz B-scan performed with reflection geometry can provide visual information about the thickness and location of defects along the depth direction [25]. Meanwhile, the THz C-scan can provide information about the surface features of the sample, including the reflectivity of the GFRP surface and the surface roughness.

In this study, defect visualization was performed in both the time and frequency domains. The visualization algorithm in the time domain using the raw THz waveform was named THz pulse time-domain visualization (TTV) and that for the frequency domain was named THz pulse frequency-domain visualization (TFV), as shown in Fig. 11. The algorithm for visualization consists of the four following steps: (i) measuring the THz signal at each pixel; (ii) recording a THz signal in one line with an x and y coordinate index; (iii) reshaping the scanned THz signal to a three-dimensional (3D) matrix; (iv) slicing the plane according to the time or frequency and subsequently imaging each signal in 2D. In the study, TTV and TFV were applied to the transmission and reflection geometries in order to compare the visualized image results.

With a scan performed using transmission geometry, we only can determine the size of the defect and its location. On the other hand, when using the reflection geometry, the defect size, location, depth, and thickness information can be visualized.

5.2.1. Transmission geometry scan results

A pulsed THz spectroscopy image using transmission geometry contains more information than a typical 2D image because each pixel of the THz-TDS image contains a whole waveform in the time domain. The peak-to-peak amplitude, arrival time, or phase of the waveform is commonly used to visualize the 2D image. Fig. 12 shows the visualization result obtained with the conventional and TTV methods. The defect is not clearly visualized with the use of the conventional method. The refractive index of GFRP is 2.16 and the Teflon film is 1.4. For the GFRP containing the simulated artificial delamination, the refractive index of the simulated delamination area is the same as air, which is 1. However, the thickness of the defect is very small in the thickness of the specimen. It means that the phase difference at maximum peak point between the transmitted THz waveforms of normal GFRP and the GFRP including defects is small. These characteristics make it difficult to visualize defects in samples with uniform thickness and material characteristics. The TTV algorithm exhibits better results than the existing visualization method (Fig. 12) because the TTV algorithm visualizes a defect image using magnitude and phase information at a specific time via a 3D data matrix.

We also performed defect visualization using the TFV algorithm to compare its performance with that of the TTV algorithm. THz C-scans in the frequency domain can also be obtained by acquiring the fast Fourier transform (FFT) of the raw waveform at each pixel. Fig. 13 shows the obtained representative visualization image based on magnitude, phase, and real-component and imaginary-component values using the TFV algorithm. We selected frequency values of 0.112, 0.225, 0.399, and 0.862 THz to compare the visualization results as a function of the frequency. Lower frequencies, corresponding to longer wavelengths, are useful for observing large scale features such as delamination, the absence of resin areas, or impact effects in composite structures. On the other hand, higher frequencies, corresponding to shorter wavelengths,

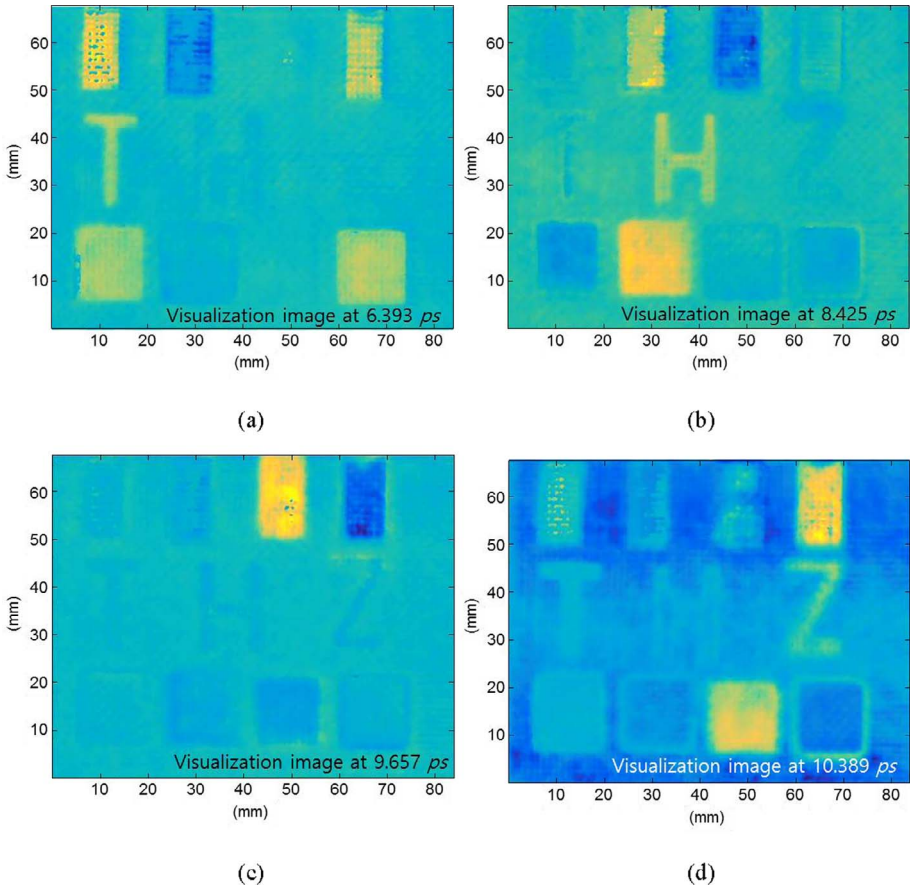

Fig. 17. THz reflection C-scan images obtained using THz time-domain visualization (TTV) after applying peak time shift algorithm according to optical time delay: (a) 6.393 *ps*; (b) 8.425 *ps*; (c) 9.657 *ps*; (d) 10.389 *ps*.

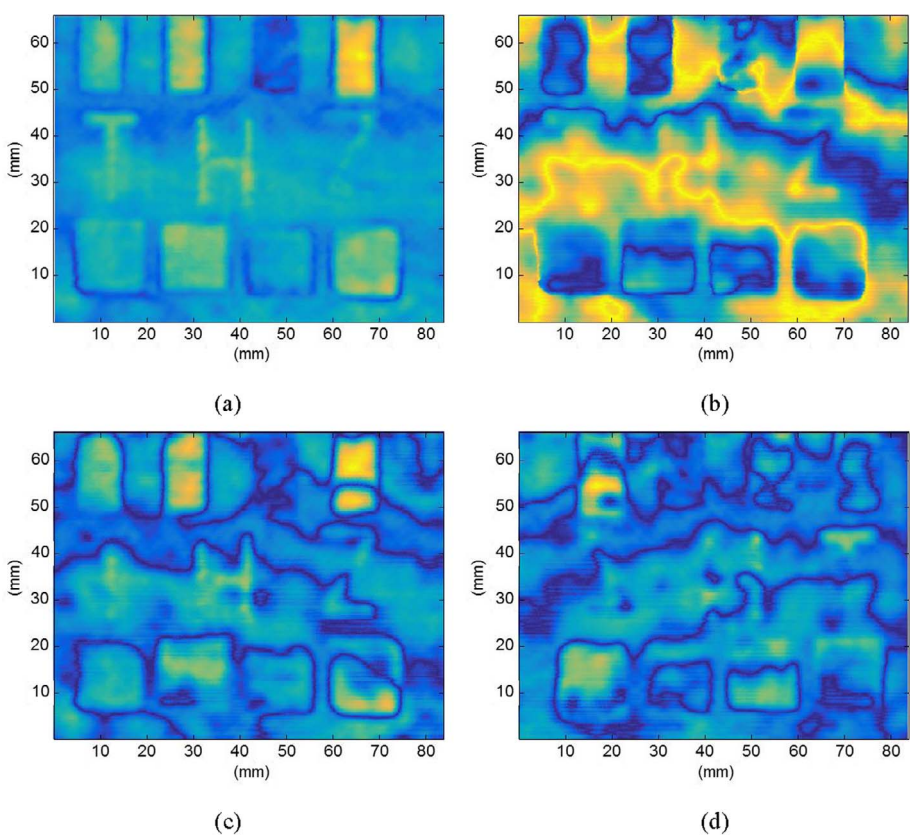

Fig. 18. THz reflection C-scan visualization images obtained using THz frequency-domain visualization (TFV): (a) magnitude map; (b) phase map; (c) real-component map; (d) imaginary-component map.

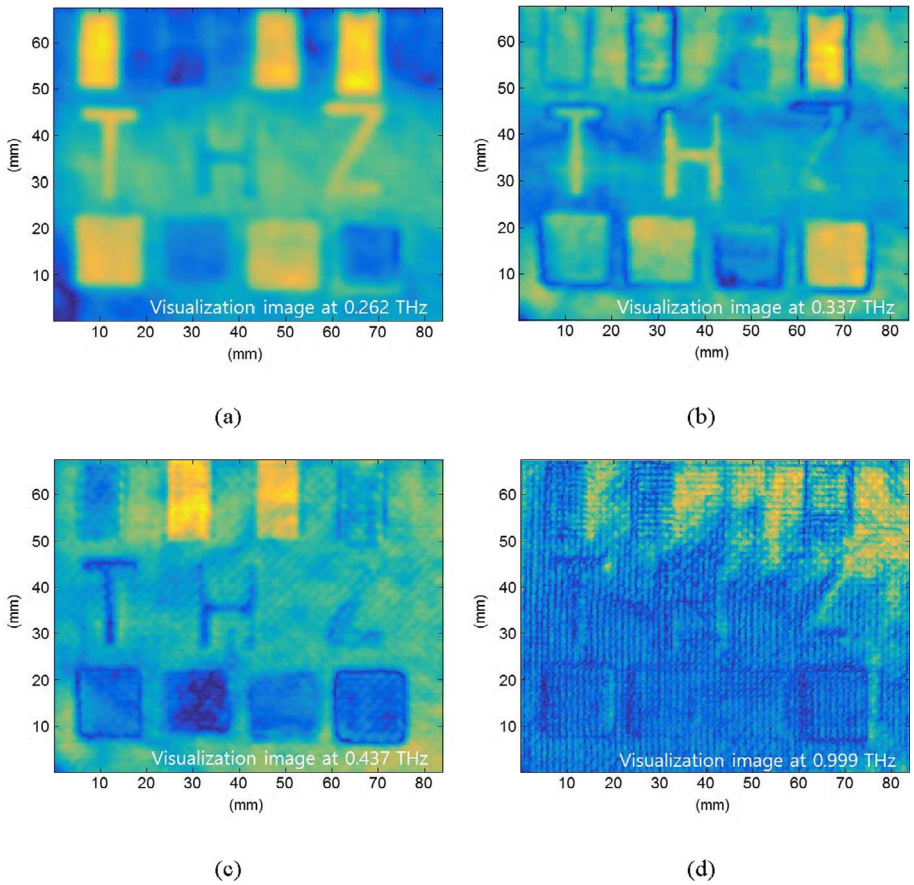

Fig. 19. THz C-scan visualization images obtained with reflection geometry at different frequencies: (a) 0.262 THz; (b) 0.337 THz; (c) 0.437 THz; (d) 0.999 THz.

are useful to visualize small cracks, fiber orientation directions, or fine shapes. The visualized images of the GFRP plate including the defect are shown in Fig. 14(a) to (d). At low frequencies, the simulated delamination and Teflon film are clearly visualized. At high frequencies, the fiber orientation appears clear. From the imaging results in Fig. 14, we note that the object of interest can be clearly distinguished even until 1.0 THz, which means that signals up to 1.0 THz can be applied for visualization. Here, we note that 1.0 THz corresponds to a wavelength of 0.3 mm.

5.2.2. Reflection geometry scan results

The THz waveform captured by using reflection geometry contains more information than that with transmission geometry, because a THz pulse that is incident on Teflon film or a simulated delamination leads to a higher difference of the reflected amplitude in the relevant time domain. These yields clearer visualization results in the B-scan image, as shown in Fig. 15.

The B-scan images in Fig. 15 also provide depth and thickness information. The narrow yellow band around the time delay of 16 ps indicates the top surface of GFRP. Regardless of defect types, the localized narrow blue band indicates the top surface of the defects and the yellow band indicates the bottom surface of the defects. In the case of the simulated multi-delamination and inserted Teflon films, their positions and thicknesses were clearly visualized when compared with the single-layer defect visualization results.

We confirmed the possibility of detecting an actual delamination by visualizing the sample at 1 mm distance from artificially created delamination boundaries. The cross section line for visualization (B-scan) lies outside of the Teflon film. However, it can be said that delamination occurred due to the thickness of the Teflon film during the fabrication process. Fig. 16 shows the possibility of visualizing an actual delamination.

When the GFRP sample is slightly curved or twisted due to various reasons, the captured THz waveform can be shifted by the geometry shape of the sample. This means that even if multiple defects exist in the same layer, some of the defects can appear to exist at different layers upon construction of the C-scan image. The captured THz waveforms can be aligned to compensate for the time delay based on the first peak time of the reference THz waveform. Fig. 17 shows the C-scan image as a function of the optical delay time using the TTV algorithm after applying the peak time shift algorithm.

Fig. 17(a) depicts a representative image between the first and second layers of the GFRP, showing a simulated delamination with a rectangular shape and the inserted Teflon film in the shape of a square and the letter T. Both simulated defects between layers two and three and three and four were successfully visualized. Defect visualization using the TFV algorithm was also performed for the purpose of comparison with the TTV result. Fig. 18 illustrates the representative visualization image based on the magnitude, phase, and real and imaginary components with the use of the TFV algorithm. The defect visualization results based on the phase difference show the boundary of the defect more clearly than in the case of the “magnitude image”. Fig. 19 shows the visualization image result according to the frequency, set to 0.262, 0.337, 0.437, and 0.999 THz. As explained in 5.2.1, lower frequencies, corresponding to longer wavelengths, show large features such as the simulated delamination and Teflon film in the GFRP structure.

6. Conclusion

In this study, we performed THz imaging of GFRP composites, including simulated rectangular delamination, square-shaped Teflon film, and Teflon film spelling “THZ,” in both the time and frequency domains based on reflection and transmission geometries. Using a 150 μm-thick

Teflon film, we simulated a single delamination between layers 1–2, 2–3, and 3–4, and a multi-delamination was simulated between layers 1–2 and 3–4 at the same position on the x-y plane. THz imaging for detecting the multi-delamination and Teflon film in the GFRP plate was successfully performed without prior knowledge of the defect presence or location. In addition to detecting the simulated delamination and Teflon film in the C-scan, the quantitative thickness and depth information were calculated by analyzing the THz waveforms acquired by the reflection geometry scan method in the time domain. Consequently, the calculated thickness of the GFRP plate and measured thickness of the GFRP plate exhibited a close consistency (< 3.7% error), and this information was satisfactorily visualized in the B-scan image.

Comparisons between THz imaging based on TTV and that based on TFV were also carried out for application in relevant fields. The TTV algorithm was well-suited for the purpose of analyzing the thickness and determining the location of delamination in the reflection geometry scan method. However, the TFV algorithm is suitable for objects of assorted sizes to be visualized. For example, small-sized defects can be visualized at high frequencies, while larger-size defects can be efficiently visualized at low frequencies.

THz imaging may be more convenient for use in the area of NDT and allied fields, because it does not require a coupling medium. Further, THz imaging provides a higher lateral resolution than conventional techniques while evaluating the size, thickness, and location of multi-layer separation despite high attenuation of the incident signal in glass fiber or other composite structures. We believe that THz imaging can provide a nondestructive, noncontact, and nonionizing method to evaluate insulators or semiconducting materials.

Acknowledgments

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